SR2b · RESPONSE-SPECTRUM-3D-BRACED-FRAME

Response-spectrum analysis of a three-dimensional L-shaped braced frame

Brace + rotational-mass + tabulated-spectrum RSA verified on an L-shaped three-dimensional braced frame

PASS

error +0.142% · tol 1.5%

§2 TECHNICAL FEATURES TESTED

- **3-D response-spectrum analysis (RSA)**
- **Braces / axial-only members** (STRIX engine linear truss bars)
- Rigid floor diaphragms (Ux, Uy, Rz)
- **Rotational mass (MMI)** — offset-mass condensation
- Tabulated El-Centro spectrum · 5% damping
- **CQC · SRSS · ABS** · axial-force recovery

§3 PROBLEM DESCRIPTION

SR2b takes the 3-D response-spectrum pipeline onto the **brace surface** and adds **rotational mass**. Adopted as **CSI SAP2000 Verification Example 1-025** with the independent solution from **Peterson (1981)**, a **three-story, L-shaped building** is framed by **four identical two-bay X-braced frames** whose columns and diagonals carry **axial force only** — every member is a **linear two-node truss bar**. All mass is concentrated at a **centre-of-mass joint** per level with X and Y translational mass *and* a **rotational mass moment of inertia (MMI)** about Z; the plan is L-shaped, so the CM is offset from the centre of rigidity and the response is **translation–torsion coupled**. The excitation is the **El-Centro-derived tabulated response spectrum** in X with 5% damping, combined by **CQC, SRSS and ABS**; only the first two modes drive the RSA, run in the app's order on the STRIX engine (modal solve, then the RSA run reusing those eigen results). The production mass path injects translational mass only (downward `Fz = −m·g` through Loads-to-Masses), so the MMI is reproduced **physically** by four offset translational masses on the diaphragm — the STRIX engine's rigid-diaphragm transformation ($M_{red} = T^T M T$) turns them into the exact Z rotational inertia, with no production-code change. The reference is reproduced directly at its own discretisation (84 truss members, 3 concentrated story masses).

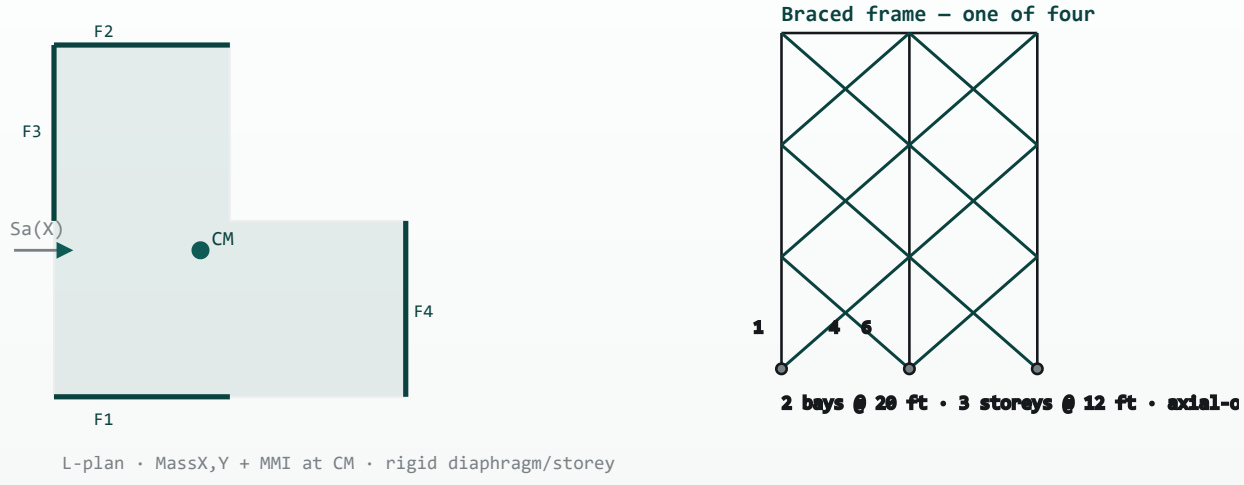


Figure SR2b. L-shaped plan with the four axial-only braced frames and the eccentric centre of mass carrying MassX, MassY and MMI (left), and the elevation of one two-bay X-braced frame with pinned, axial-only members (right). Frame-1 elements 1 (column), 4 and 6 (diagonals) are the axial-force comparison points.

§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		MEMBERS & MASS		LOADING & DYNAMICS	
Braced frames	4	E	29500 ksi	Excitation	El-Centro spectrum (X)
Bays / frame	2	Area	6 in²	Damping	5 %
Storeys	3	Members	Axial-only (truss bar)	Modes used	2
Bay / storey	20 / 12 ft	Mass	MassX,Y + MMI at CM	Combination	CQC · SRSS · ABS

§5 REFERENCE SOLUTION

Frequency — mode 1: $(K - \omega^2 M)\cdot\phi = 0$, $f = \omega/2\pi$ = 3.0592 Hz

Roof CM Ux — CQC: $u = \sqrt{\Sigma_i \Sigma_j \rho_{ij}\cdot u_i \cdot u_j}$, ρ = Der Kiureghian = 1.0329 in

Axial elm 1 — CQC: $N = \sqrt{\Sigma_i \Sigma_j \rho_{ij}\cdot N_i \cdot N_j}$ = 279.48 kip

CSI SAP2000 Software Verification, Example 1-025 (Response-Spectrum Analysis of a Three-Dimensional Braced Frame); independent solution = Peterson (1981), reproduced exactly (0 %) by SAP2000. A three-story, L-shaped building framed by four identical two-bay, three-story X-braced planar frames whose columns and diagonals carry axial force only (all pinned). The frames are tied by a rigid floor diaphragm at each level. All mass is concentrated at a centre-of-mass joint per level with X and Y translational mass (1.24224 kip-s²/in) and a rotational mass moment of inertia about Z (174,907.4 kip-in-s²), giving nine dynamic DOF; for consistency with Peterson only the first two modes drive the RSA. The excitation is the El-Centro-derived 5%-damped response spectrum applied in X, combined by CQC, SRSS and ABS. Reference values are the two modal frequencies, the roof centre-of-mass (joint 51) X/Y displacement and Z rotation, and the axial forces in three Frame-1 elements (a column and two diagonals) under each combination rule.

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Frequency — mode 1 (Hz)	eigen · Hz	3.05916	3.0592	−0.0010%	PASS
Frequency — mode 2 (Hz)	eigen · Hz	3.11876	3.1188	−0.0010%	PASS
Roof CM Ux — CQC (in)	CQC · in	1.03204	1.0329	−0.083%	PASS
Roof CM Ux — SRSS (in)	SRSS · in	0.736576	0.7372	−0.085%	PASS
Roof CM Ux — ABS (in)	ABS · in	1.04144	1.0423	−0.083%	PASS
Roof CM Uy — CQC (in)	CQC · in	0.141323	0.1414	−0.054%	PASS
Roof CM Uy — SRSS (in)	SRSS · in	0.736576	0.7372	−0.085%	PASS
Roof CM Uy — ABS (in)	ABS · in	1.04144	1.0423	−0.083%	PASS
Roof CM Rz — CQC (rad)	CQC · rad	2.5164×10^{-4}	2.52×10^{-4}	−0.142%	PASS
Roof CM Rz — SRSS (rad)	SRSS · rad	2.5164×10^{-4}	2.52×10^{-4}	−0.142%	PASS
Roof CM Rz — ABS (rad)	ABS · rad	2.5164×10^{-4}	2.52×10^{-4}	−0.142%	PASS
Axial elm 1 — CQC (kip)	CQC · kip	279.249	279.48	−0.082%	PASS
Axial elm 1 — SRSS (kip)	SRSS · kip	200.385	200.55	−0.082%	PASS
Axial elm 1 — ABS (kip)	ABS · kip	281.763	281.99	−0.081%	PASS
Axial elm 4 — CQC (kip)	CQC · kip	194.342	194.5	−0.081%	PASS
Axial elm 4 — SRSS (kip)	SRSS · kip	139.457	139.57	−0.081%	PASS
Axial elm 4 — ABS (kip)	ABS · kip	196.091	196.25	−0.081%	PASS
Axial elm 6 — CQC (kip)	CQC · kip	120.423	120.52	−0.081%	PASS
Axial elm 6 — SRSS (kip)	SRSS · kip	86.4136	86.48	−0.077%	PASS
Axial elm 6 — ABS (kip)	ABS · kip	121.507	121.61	−0.085%	PASS

§7 MODAL COMBINATION

METHOD	ROOF CM UX (IN)	DIFF
CQC	1.03204	−0.083%
SRSS	0.736576	−0.085%
ABS	1.04144	−0.083%

The reference publishes the roof centre-of-mass displacement under **three modal-combination rules**, and STRIX reproduces each: **CQC** and **SRSS** through the production RsaCalculator (Der Kiureghian CQC), and **ABS** (absolute sum) from the same per-mode responses. Because the two modes are **closely spaced** (frequencies within 2 %), STRIX's CQC (`1.0320 in`) far exceeds its SRSS (`0.7366 in`) — the cross-correlation adds the modes nearly in phase in X — while ABS (`1.0414 in`) is the upper bound. All three land `within tolerance` of the independent reference (1.0329 / 0.7372 / 1.0423 in; about −0.08% each), verifying that STRIX handles closely-spaced modes correctly (the very case that motivates CQC).

§8 CONCLUSION

PASS

STRIX reproduces the **two modal frequencies**, the **roof centre-of-mass X/Y/Rz response**, and three **member axial forces** of the **CSI 1-025 / Peterson 1981** L-shaped braced frame under **all three combination rules** to **better than a quarter of a percent** across twenty independent quantities (worst 0.142%, roof Rz). This verifies three surfaces at once: the **brace / axial-only member** in the modal and RSA path (linear truss bar), a **rotational mass moment of inertia** reproduced through the rigid-diaphragm mass condensation (offset translational masses → exact Rz inertia), and a **tabulated spectrum with closely-spaced modes**. Together with SR1 and SM5/SM5b/SM6 it extends STRIX's response-spectrum verification from planar frames to a braced 3-D configuration.

§9 REFERENCES & ANALYSIS FILES

1. CSI SAP2000 Software Verification, Example 1-025 — Response-Spectrum Analysis of a Three-Dimensional Braced Frame.
2. F. E. Peterson (1981) — independent response-spectrum solution reproduced exactly by SAP2000.
3. E. L. Wilson, A. Der Kiureghian & E. P. Bayo, *EESD* **9**(2), 1981 — CQC rule. STRIX harness: `buildEigenPy → STRIX engine (eigen route; rigid-diaphragm transformation, M_red = TᵀMT) → buildRsaRunPy (reuses the eigen results) → STRIX engine (rsa route) → rsa_modal.bin`; MMI via four offset translational masses on the rigid diaphragm.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · SR2b/` `sha256 (pending publish)`